

A Topological Loss Function for Deep-Learning Based Image Segmentation Using Persistent Homology, IEEE TPAMI

1. Place of origin

The article is printed in the international journal IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI), published by the IEEE Computer Society. Has Q1 quartile.

2. Time of origin

The article is dated december 2022 (volume 44, number 12).

3. Author

The article is written by a group of authors - James Clough, Nicholas Byrne, Ilkay Oksuz, Veronika Zimmer, Julia Schnabel, and Andrew King, mostly affiliated in London and one from Istanbul and one from German.

4. Theme / Topic

The article deals with training convolutional neural networks (CNN) for image and volume segmentation. It addresses the problem that standard segmentation loss functions ignore the large-scale shape, or topology, of the segmented object, and proposes a way to incorporate prior topological knowledge into training using persistent homology, a concept from topological data analysis.

5. Main idea / Aim of the article

The main idea is that pixel-wise loss functions, such as cross-entropy or the Dice loss, only measure overlap between the predicted and ground-truth segmentations and cannot capture whether its large-scale structure is topologically correct. The aim of the authors is to provide a differentiable topological loss function that specifies the desired Betti numbers of an object - the number of connected components, loops, and cavities - and to show

that driving the network towards the correct topology also improves pixel-wise accuracy, even without ground-truth labels for every case.

6. Contents of the article

The article can be divided into four parts. The first part deals with the introduction and related work, explaining why pixel-wise losses fail to capture topology. The second covers the theory: persistent homology of cubical complexes, Betti numbers, barcode diagrams, and the topological loss function L_{topo} , applied either as post-processing or in a semi-supervised framework. The third touches upon four experiments: denoising MNIST digits; segmenting the hearts myocardium from MRI data; the public ACDC challenge dataset; and segmenting the placenta from 3-D ultrasound. The fourth part includes the discussion and conclusion, addressing the methods limitations and further applications.

The article is written mostly in the first person plural (like :we) and includes equations, result tables, and figures such as barcode diagrams. The authors start by stating that pixel-wise losses cannot capture topological correctness, then describe the theory and derive the loss function. According to the text, across all four experiments the topological loss consistently improved both the Dice score and the proportion of topologically correct segmentations, especially on harder tasks. The authors conclude that persistent homology is a viable tool for embedding topological prior knowledge into segmentation training.

7. Personal opinion / impression of the article

I found the article interesting and useful, since it presents a genuinely new approach to combining topology with machine learning. Because topology is, by its nature, concerned with transformations of shapes and connectivity patterns, it seems logical that it finds application in computer vision, and in segmentation in particular. I believe this direction - applying persistent homology to segmentation - is very likely to become the topic of my own masters thesis, aiming at either faster inference, faster training, or better accuracy than classical pixel-wise loss functions.

8. Personal view on the topic / idea / problem

I share the authors view that pixel-wise loss functions alone are insufficient whenever the global shape of an object matters. At the same time, I see the problem in a slightly different way: computing persistent homology adds computational cost, on the order of seconds per batch or per volume, which makes the method unsuitable for real-time segmentation. To my mind, this is the main bottleneck of the approach, and improving the speed of the computation is the most important open direction for future work, including a possible direction for my own research. I don't quite agree that computational optimisation should be left outside the scope of the paper: without it, topological loss functions are unlikely to be widely adopted in real clinical or industrial pipelines.